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High-precision services of BeiDou navigation satellite system (BDS): current state, achievements, and future directions

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Abstract

With the rapid development of technologies such as autonomous driving and robotic navigation, the demand for accurate and reliable positioning results with BDS has increased. The current status and future development of BDS high precision services are reviewed. The operational PPP-B2b of BDS-3 achieves positioning accuracy at decimeter-level within 14 min and has the drawbacks in regional coverage and long convergence time compared with the international counterparts, such as CLAS of QZSS and HAS of Galileo. A development frame for multi-layer BDS high-precision services is proposed by considering its construction cost, compatibility of user terminal, operating maintenance burden, and service level. The global high-precision service with the improved orbit and clock accuracy enhanced with the inter-satellite link is taken as the most fundamental layer. By incorporating inter-satellite link observations, the orbit errors URE and clock errors are significantly reduced to approximately 0.05 m and 0.16 ns, respectively, which will expand PPP service to global. Based on this, the regional PPP-AR and PPP-RTK services with improved convergence performance are taken as the second layer. With PPP-AR, the convergence time drops to 10 min. With PPP-RTK, the convergence time would be further shortened to less than 3 min. The LEO enhanced PPP service is taken as the third layer which overcomes both drawbacks of the long convergence time of the inter-satellite link enhanced service (the first layer) and the regional coverage of the PPP-AR/PPP-RTK services (the second layer). The simulation results show that by introducing a LEO constellation of 288 satellites, the LEO enhanced PPP can achieve positioning accuracy better than 5 cm within approximately 1 min. In addition, the integration of LEO constellation and PPP-RTK is further proposed to enable instantaneous convergence. The implementation challenges are also presented.

Keywords BeiDou navigation satellite system, Precise point positioning, PPP-B2b, Low-earth-orbit, PPP-RTK, PPP-AR, Orbit, Clock

Introduction

As a Chinese fundamental facility in space and timing information transfer, the development of Beidou Satellite Navigation System (BDS) follows three steps. On July 31, 2020 BDS-3 constellation including 3 GEO satellites, 3 IGSO satellites, and 24 MEO satellites was completed and started providing global services, including positioning navigation and timing services and six featured services. Currently, four systems compete on the same stage including modernized Global Positioning System (GPS), GLObal Navigation Satellite System (GLONASS), BDS-3, and Galileo (Li et al., 2014a, 2014b, 2022b; Yang et al.,

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2020, 2022). As evaluated by Montenbruck et al. (2020), the Signal-in-Space Error (SISRE) of BDS, Galileo and GPS broadcast messages all are less than 1 m.

The development of economic activities derives fast and high-precision positioning services for mass-market applications (European GNSS Agency, 2019; Li et al., 2022a). Obviously, the accuracy of BDS Single Point Positioning (SPP) is only at meter-level (Chen et al., 2020) and cannot meet the demands of precise applications. Therefore, augmentation systems have been constructed to improve the accuracy, integrity, continuity, and availability of GNSS performance by broadcasting additional corrections to users (Li et al., 2018, 2020a, 2020b). The augmentation systems are divided into Ground-Based Augmentation System (GBAS) and Satellite-Based Augmentation System (SBAS) according to the broadcasting medium of augmented information (Yang et al., 2022). GBAS broadcasts the orbit and clock corrections and integrity parameters of navigation satellites to users through terrestrial radio or Internet and enables precision approach guidance (Felux et al., 2015). In 2003, Ordnance Survey (OS) Net was established as a national network of nearly 100 GPS base stations, which constantly transmit GPS-observed positioning data from UK (Nabhan et al., 2008). To provide more accurate corrections over broader areas (Hu et al., 2003), CORSnet-NSW began providing Continuously Operating Reference Stations (CORS) services in Sydney, New South Wales over a decade ago (Janssen & Haasdyk, 2011; McMahon et al., 2022). China also developed the National BeiDou Ground-based Augmentation System (NBGAS), which can provide real-time positioning services at the centimeter level accuracy and post-processing positioning with millimeter level (Li et al., 2020a, 2020b). Although GBAS provides services with high accuracy and integrity, it involves active communication of users and frequent interaction between users and the operation center. Unlike GBAS, which broadcasts augmentation corrections through ground-based network, SBAS broadcasts augmentation corrections via satellites. SBAS systems broadcast orbit and clock corrections and integrity parameters to users through GEO satellites, which improves the accuracy and integrity of GNSS services (Zheng et al., 2022). The typical SBAS systems include the Wide Area Augmentation System (WAAS) in the United States, the System of Differential Correction and Monitoring (SDCM) in Russia, the European Geostationary Navigation Overlay System (EGNOS) in Europe, the BeiDou Satellite-Based Augmentation System (BDS-BAS) in China, the Mtsat Satellite-based Augmentation System (MSAS) in Japan, and the GPS Aided GEO Augmented Navigation (GAGAN) in India (Shao et al., 2020). The above-mentioned typical SBAS systems provide civil

aviation users at CAT-1 phase with meter-level positioning accuracy.

Recently, another development is that GNSS constellations directly broadcast centimeter accuracy SSR corrections for PPP services free of charge (Liu et al., 2020). The new augmentation service uses the signals broadcast by the GNSS system itself to carry augmentation corrections, rather than relying on additional ground-based networks or communications satellites like GBAS or SBAS. The Centimeter-Level Augmentation Service (CLAS) is carried out on the L6D signal of Japanese Quasi-Zenith Satellite System (QZSS) and able to reach centimeter-level accuracy within 1 min in QZSS coverage around Japan within the PPP-RTK (Real-Time Kinematic) frame (Miya et al., 2016; Cabinet Office, 2018; Satoshi Kogure, 2021). Besides, QZSS also uses L6E signal to provide precise positioning services to users in the Asia-Pacific region with centimeter accuracy (Li & Chen, 2017). In Russia, GLONASS announces its PPP service plan, which is expected to be used in Emergency Road Service (ERS), unmanned driving, and precision engineering with an accuracy of 0.1 m (Revnivkykh, 2019). The Galileo High Accuracy Service (HAS) is developed by the European Union and provides freely precise satellite orbit, clock, DCB, and UPD parameters of GPS L1/L5/L2C signals and Galileo E1/E5b/E5a/E6/E5 signals worldwide through E6 signal of MEO satellites (EUSpace, 2023). According to an assessment research, the RMS for the orbit Signal-In-Space Ranging Error (SISRE) is 1.6 cm and 1.4 cm for Galileo and GPS, respectively, while the RMS of the total SISRE is 10.6 cm and 11.8 cm, respectively. In addition to orbit and clock corrections, satellite code bias corrections also have a good consistency with the final code biases from DLR, with RMS of 0.29 ns, 0.27 ns, and 0.23 ns for Galileo C1C–C5Q, C1C–C7Q, and C1C–C6C, respectively, and 0.21 ns for GPS C1C–C2L (Naciri et al., 2023). The Galileo HAS is expected to provide precise real-time positioning services with horizontal positioning error at 95% better than 20 cm and vertical positioning error at 95% better than 40 cm and the convergence time globally less than 5 min and shorter in Europe. The results and preliminary analysis of the HAS initial service indicate that the HAS full-service targets are feasible (Fernandez-Hernandez et al., 2022).

BDS-3 also provides precise corrections through PPP-B2b signals of GEO satellites (CSNO, 2019a, 2019b; Tang et al., 2022; Yang et al., 2021). With the code and phase observations at the ground stations, the operation control center generates the precise orbit and clock corrections of BDS-3 IGSO/MEO and GPS satellites and sends them to the GEO satellites. Then, the corrections are broadcasted to the users in China and surrounding areas through PPP-B2b signals (Yang et al., 2020, 2022).

The PPP-B2b signal broadcasts the I-component and the Q-component with a center frequency of 1207.14 MHz and a bandwidth of 20.46 MHz, and the data modulation of the signal is Binary Phase Shift Keying (BPSK) and the message data rate is 500 bit/s. The basic frame structure of the PPP-B2b_I navigation message is described in BeiDou Navigation Satellite System Signal In Space Interface Control Document Precise Point Positioning Service Signal PPP-B2b, and users can decode the augmentation corrections according to the definition in the document. With PPP-B2b corrections, decimeter-level kinematic positioning accuracy and centimeter-level static positioning accuracy are expected. (Xu et al., 2021; Yang et al., 2022).

Emerging applications, such as autonomous driving, unmanned aerial vehicles, and smart device navigation, require high-precision positioning services, which the current BDS-3 PPP-B2b service is unable to meet because of its regional coverage and longer convergence time of more than ten minutes. The BeiDou high-precision navigation service is committed to improving its service performance in terms of convergence time and coverage. Under the guidance of the concepts of comprehensive, resilient, and secure PNT (Positioning, Navigation, and Timing), Yang et al. (2023) have emphasized the importance of establishing a high-precision Chinese spatiotemporal datum to meet the demands of high-precision navigation and positioning. Furthermore, they pointed out that BDS inter-satellite link (ISL) technology and emerging low Earth orbit (LEO) navigation augmentation technology are expected to enhance the performance of the Beidou high-precision services (Yang and Ming, 2023; Yang et al., 2024). Although some articles such as by Ge et al. (2022), Geng et al. (2011), Li et al. (2020a, 2020b), and Li et al. (2021) presented the proposals on improving precise point positioning performance of BDS, no methodologies are given on how to implement the proposals within the frame of BDS-3 system. Different from the previous studies, this contribution focuses on how to implement high precision service performances within the frame of BDS-3 system.

This contribution first summarizes the current state of BeiDou high-precision navigation service. Then the advancements improving high-precision performance within the BDS-3 frame are provided. LEO constellation enhanced BDS-3 high-precision services in terms of coverage and convergence are proposed. In “[Current state of BDS high-precision services](#)” section, we introduce the current performance indicators of BeiDou PPP-B2b services and compare them with mainstream precise navigation augmentation services worldwide. An analysis will be conducted to identify the metrics that need improvement in PPP-B2b services. In “[Key technological](#)

[advancements in BDS](#)” section presents the recent technical advancements which can be realized in the BDS-3 frame for improving the high precision performance. In “[Future directions of BDS high-precision services](#)” section gives various enhancement proposals for the future development based on the LEO constellation. Finally, the proposed solutions will be summarized, along with an outlook on the future development trends, potential increments, and bottleneck issues in the high-precision services offered by BeiDou.

Current state of BDS high-precision services

BDS B2b service

PPP-B2b is the only high precision service providing decimeter level positioning accuracy among the designed BDS-3 services. PPP-B2b has been available to users since July 31, 2020. The precise orbit and clock corrections and integrity parameters for GPS and BDS-3 IGSO and MEO satellites are broadcasted to users in China through PPP-B2b signals of GEO satellites. (Liu et al., 2020; Tang et al., 2022).

The PPP-B2b signal of GEO satellites broadcasts both slow-varying augmentation information and fast-varying augmentation information. The slow-varying augmentation information is updated every 48 s, including PRN Mask, Differential Code Bias, and Orbit Correction, and User Range Accuracy Index. The fast-varying augmentation information, which refers to Clock Corrections, is updated every 6 s. The broadcast information type as well as the update interval is listed in Table 1.

The precise orbit corrections refer to the phase-center of the satellite rather than the center-of-mass. Users should no longer conduct the satellite phase center offset corrections. The PPP-B2b clock corrections refer to the phase center of L1/L2 combinations for GPS and B3I for BDS-3 satellites. Users should use the PPP-B2b broadcast DCB parameters to account for the signal-dependent delay differences. The correction approaches of the DCB parameters are in the same way as those of the broadcasted messages.

Table 1 Information types of PPP-B2b message

Type No	Type	Update interval (s)	Nominal validity (s)
Type 1	PRN mask	48	–
Type 2	Orbit correction and user range accuracy index	48	96
Type 3	Differential code bias	48	86,400
Type 4	Clock correction	6	12

The PPP-B2b broadcast orbit and clock solutions are compared with IGS GFZ precise products for accuracy assessment. The orbit accuracy and clock accuracy are assessed separately. URE in Table 2 refers to the orbit-only URE. From Table 2, it can be concluded that GPS and BDS-3 orbit with centimeter level accuracy URE and clock solutions with 0.2 ns accuracy can be obtained with PPP-B2b broadcast information.

Six monitoring stations within China from March 22nd to March 29th 2022 are used to evaluate positioning performance of PPP-B2b services. Kinematic positioning experiments are conducted. Detailed positioning strategy is summarized in Table 3.

The convergence time is evaluated and given in Fig. 1. Convergence is defined when the horizontal positioning error is smaller than 30 cm and vertical positioning error is smaller than 60 cm for at least 5 min. The average convergent time is about 14 min for BDS-3 only positioning, while it drops to less 10 min when GPS is added.

The position error at 95% confidential level after one-hour convergence is calculated and given in Fig. 2. For BDS-3 only positioning, the horizontal positioning error (at 95%) is better than 15 cm and the vertical positioning error (at 95%) is better than 25 cm. For BDS/GPS dual-mode positioning, the horizontal positioning error

(at 95%) is better than 10 cm and the vertical positioning error (at 95%) is better than 20 cm.

Comparison of BDS B2b, Galileo HAS, and QZSS

Besides PPP-B2b of BDS-3, Galileo and QZSS also provide precise positioning services. The details of the three types of precise positioning services are shown in Table 4. Compared with Galileo HAS and QZSS precise positioning services, the longer convergence time and regional coverage of PPP-B2b are noticed. Several reasons account for the longer convergence time and regional coverage. Firstly, precise corrections are broadcasted through GEO PPP-B2b signals and only regional measurements are used in orbit and clock determination, resulting that PPP service of BDS-3 is only provided around China rather than the global. Besides, compared with QZSS CLAS which serves as PPP-RTK and Galileo HAS which serves as PPP-AR, PPP-B2b of BDS-3 provides standard PPP services. Therefore, PPP-B2b users have a longer convergence time.

The next progress of BDS precise positioning services should aim at shorter convergence time and global coverage. The following sections of this paper introduce the strategies improving the coverage and convergence performance of BDS PPP.

Table 2 RMS of PPP-B2b orbit corrections and clock corrections

Satellite type	Radial (m)	Along-track (m)	Cross-track (m)	Orbit-only URE (m)	STD of clock difference (ns)
BDS-3 IGSO	0.15	0.42	0.38	0.16	0.39
BDS-3 MEO	0.05	0.30	0.28	0.08	0.22
GPS	0.07	0.39	0.31	0.10	0.31

Table 3 Kinematic PPP strategy

Items	PPP strategy
Observations	Ionosphere-free code and phase of L1/L2, B1I/B3I
Sampling	60 s
weight	Code to phase is 1:10,000
Receiver-end PCO/PCV	Values from manufactories
Satellite-end PCO/PCV	IGS PCO/PCV for GPS, PCO Values from manufactories for BDS
Station displacement	IERS conventions 2010
Relativistic	Corrected
Troposphere delay	Saastamoinen model with GMF mapping function, the residual tropospheric delay is estimated as random-walk process
Receiver clock	Estimated as white noise
Receiver position	Estimated as white noise
Phase ambiguities	Estimated
Estimation	Kalman filter

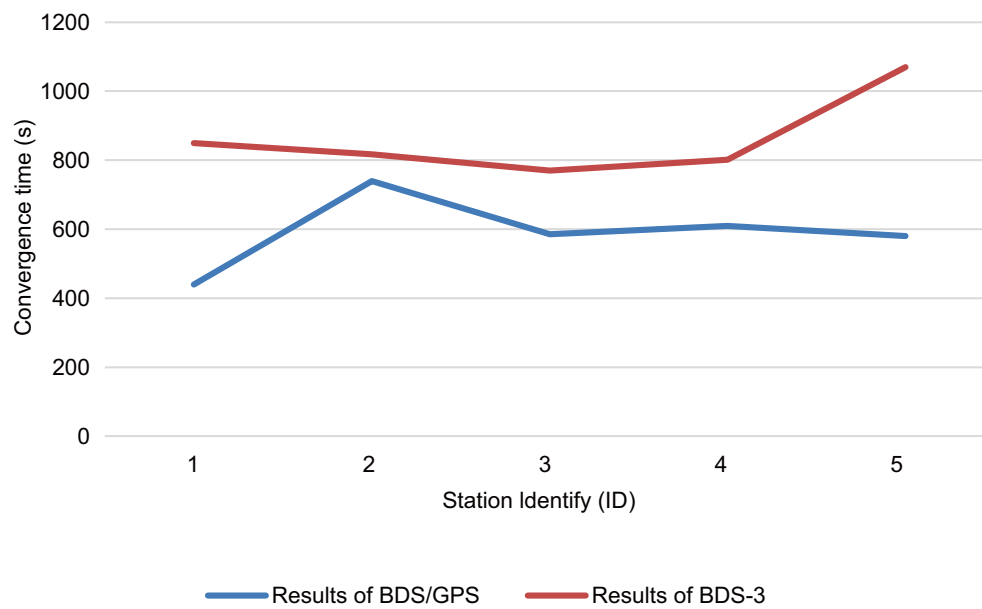


Fig. 1 Convergence Time of PPP-B2b Positioning Performance. Convergence is defined when the horizontal positioning error is smaller than 30 cm and vertical positioning error is smaller than 60 cm for at least 5 min. The average convergent time is about 14 min for BDS-3 only positioning, while it drops to less 10 min when GPS is added

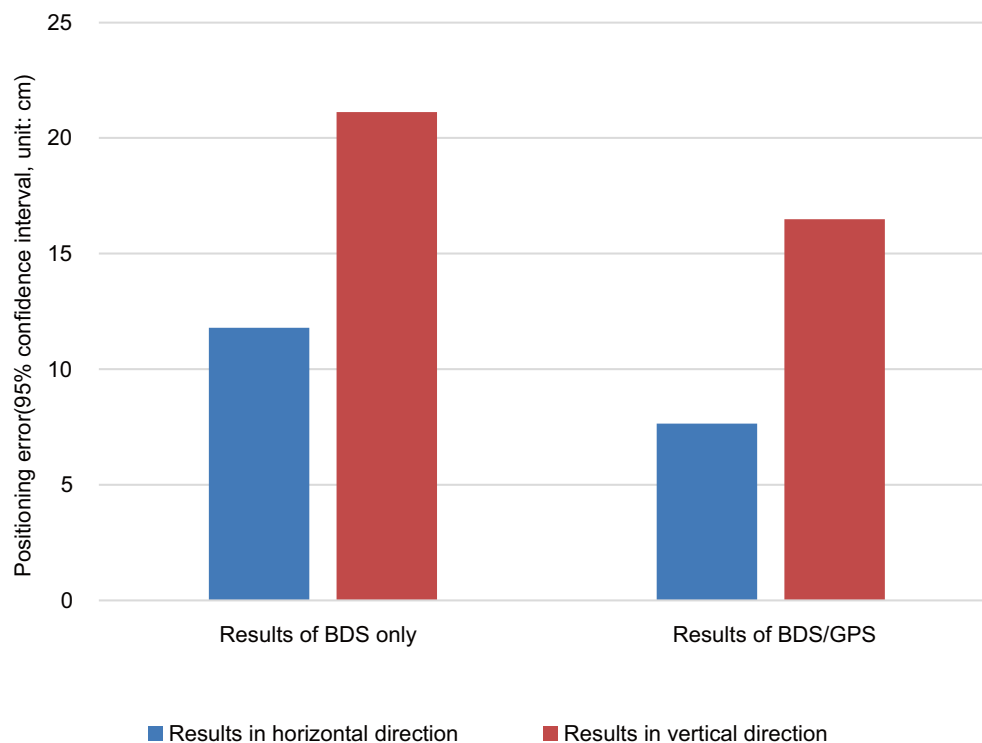


Fig. 2 The position error of 95% confidential level of PPP-B2b service. For BDS-3 only positioning, the horizontal positioning error (at 95%) is better than 15 cm and the vertical positioning error (at 95%) is better than 25 cm. For BDS/GPS dual-mode positioning, the horizontal positioning error (at 95%) is better than 10 cm and the vertical positioning error (at 95%) is better than 20 cm

Table 4 Kinematic PPP strategy

Service	BDS PPP-B2b	QZSS		Galileo HAS	
		CLAS	Fast-PPP Ex	Service level 1	Service level 2
Converge	China and surrounding areas	Japan only	Asia and Pacific Region	Global	European Coverage Area
Starting service	July 31st, 2020	November, 2018	Will be available on 2024	January 2023, Initial Service Declaration. Full Service Operational Capability will be declared on 2024	
Broadcasting channel	PPP-B2b of GEO satellites	Inclined Geosynchronous Orbit satellites 1–4 L6D	QZSS satellites 1–7 L6E, QZSS satellites 5–7 L6D	Galileo E6B	
Augmented GNSS	BDS-3 and GPS	GPS	GPS	GPS and Galileo	
Broadcasted information	Precise orbit, clock offsets and DCB parameters	Precise orbit, clock offsets, DCB parameters, phase bias, ionospheric delay and tropospheric delay	Precise orbit, clock offsets, DCB parameters, phase bias, ionospheric delay	Precise orbit, clock offsets, DCB parameters, phase bias	
Convergence time	BDS-3 only: 30 min; BDS-3 and GPS: 20 min	1 min	Not reported	300 s	
Positioning accuracy	BDS-3 Only Horizontal Positioning Accuracy (95%): 30 cm; Vertical Positioning Accuracy (95%): 60 cm; BDS-3 and GPS- Horizontal Positioning Accuracy (95%): 20 cm; Vertical Positioning Accuracy (95%): 40 cm	Static: Horizontal Positioning Accuracy (95%): 6 cm; Vertical Positioning Accuracy (95%): 12 cm; Kinematic: Horizontal Positioning Accuracy (95%): 12 cm; Vertical Positioning Accuracy (95%): 24 cm;	Not reported	Horizontal Positioning Accuracy (95%): 20 cm; Vertical Positioning Accuracy (95%): 40 cm	

Key technological advancements in BDS PPP-AR and PPP-RTK

Using the precise satellite orbit and clock products broadcasted by the BDS PPP-B2b service, users can achieve real-time float PPP solutions. However, the convergence time for float PPP solutions is typically over tens of minutes, which is unacceptable for real-time precise positioning services. By adding Uncalibrated Phase Delay (UPD), ionosphere and tropospheric delays transmitted through the GEO PPP-B2b signal along with current orbit and clock corrections, PPP-RTK service instead of PPP service can be expected. The convergence time for PPP-B2b service can be further reduced to several minutes.

Wübbena et al. (2005) first put forward the notion of PPP-RTK. It serves as an advancement of the PPP model, providing users with the ability to implement ambiguity resolution and attain absolute positioning with enhanced precision and accelerated convergence speed. In the previous decade, numerous studies focused on developing diverse PPP-RTK models and assessing performances (Geng et al., 2011; Li et al., 2011; Teunissen et al., 2010). The PPP-RTK model involves the successive generation of satellite orbit, satellite clock, uncalibrated phase bias, and atmospheric delays based on undifferenced

model. Using a multilayer processing scheme (Li et al., 2011, 2014a, 2014b), the precise results attained at one layer contribute to the generation of corrections at the next layer. Utilizing the atmospheric augmentation corrections based on more than 1000 stations in China, the PPP-RTK can implement instantly ambiguity resolution and obtain centimeter-level positioning accuracy in a few minutes in China.

In contrast to the Observable Space Representation (OSR) correction employed in RTK, which aggregates all GNSS-related errors into range corrections, the proposed PPP-B2b PPP-RTK corrections take the form of State Space Representation (SSR), offering a range of separate corrections. The newly broadcast information of PPP-B2b include ionospheric and tropospheric delays, as well as Uncalibrated Phase Delay (UPD). The framework of BDS PPP-RTK service can be designed as Fig. 3.

As shown in Fig. 3, the BDS PPP-B2b PPP-RTK service system consists of three main components: the ground reference station network, the data center, and the broadcasting system. The ground reference station network is responsible for receiving GNSS observation data and transmitting it in real-time to the data center, which utilizes these observations to generate various SSR products and sends them to the broadcasting system. The

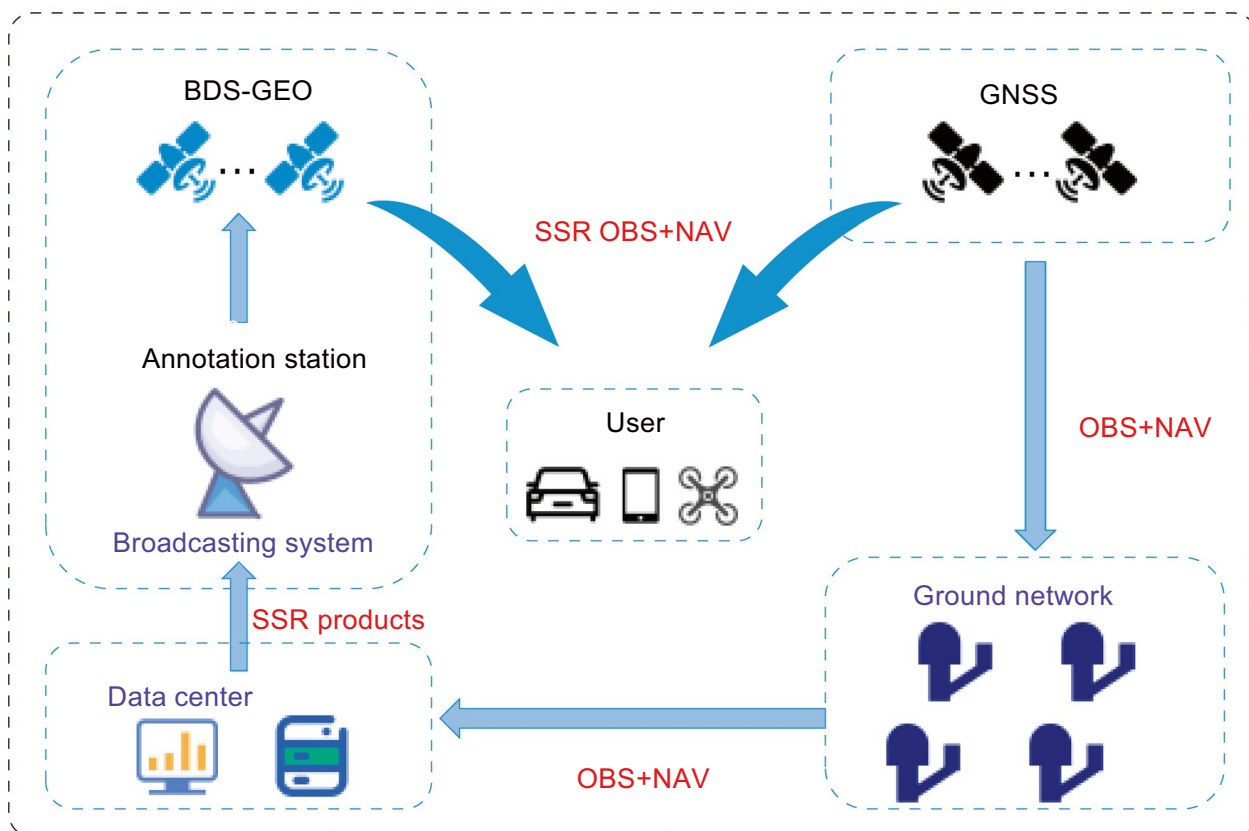


Fig. 3 BDS PPP-AR/PPP-RTK service system. The BDS PPP-B2b PPP-RTK service system consists of three main components: the ground reference station network, the data center, and the broadcasting system. The ground reference station network is responsible for receiving GNSS observation data and transmitting it in real-time to the data center, which utilizes these observations to generate various SSR products and sends them to the broadcasting system. The broadcasting system, in turn, uploads the SSR correction data to BDS GEO satellites through satellite data annotation stations and broadcasts it to users. At the user end, by combining GNSS satellite observation data with SSR correction data from BDS GEO satellites, real-time and precise PPP-RTK positioning can be achieved rapidly. When the atmospheric corrections are not available, the PPP-RTK turns into PPP-AR. The broadcast code and phase biases products can still be used to achieve PPP ambiguity resolution (PPP-AR), which can achieve higher precision and faster convergence than the float PPP solutions

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To achieve real-time and centimeter-level accuracy positioning performance, the accuracy specifications for the correction products in the service system are presented in Table 5. The code bias and UPD product are expected to achieve an accuracy of at least 0.2 ns and 0.1 cycles, respectively. The grid tropospheric and

Table 5 Products accuracy specifications of BDS PPP-AR/PPP-RTK service

Products	Interval	Accuracy
Code bias	1 h	0.2 ns
UPD	30 s	0.1 cycle
Grid tropospheric correction	30 s	2 cm
Grid ionospheric correction	5 s	5 cm

ionospheric correction products should have an accuracy of better than 2 and 5 cm, respectively. In terms of ground station distribution, to achieve PPP-AR, dozens of sparsely distributed ground stations can meet the accuracy requirements of code and phase bias products. However, to achieve PPP-RTK, a much denser ground network is required. The spacing of ground stations

should be controlled between 150 and 300 km in the areas of mid to high latitudes with stable atmospheric changes, and less than 150 km in the areas of low latitudes with severe atmospheric activities. Based on the product accuracy, Table 6 provides the performance metrics concerning the BDS PPP-AR/PPP-RTK service.

It can be concluded from Table 6 that, compared with PPP-B2b service, PPP-AR and PPP-RTK improve the convergence time of BDS-3 high-precision service greatly.

Orbit and clock accuracy improvement with inter-satellite links

Precise orbit determination and clock estimation for BDS-3 precise services rely on a regional ground monitoring network. Generation of precise orbit and clock offsets for BDS-3 satellites based on the regional sparsely distributed ground monitoring stations is the most important prerequisite for global precise positioning service. The inter-satellite link was once used to enhance accuracy of the broadcast orbit and clock offsets for BDS-3 satellites. This section proposes an approach for the real time determination of orbit and clock offsets with centimeter accuracy for global BDS-3 high-precision service by combining inter-satellite link measurements and the observations at regional ground spare monitoring stations. The strategy comprises of two processes. The first one is real time orbit determination to generate precise orbits and then a new precise clock offsets determination approach is followed using inter-satellite link measurements only. The strategy is organized as follows. An orbit determination process is conducted every hour by combining code and phase measurements from regional network and inter-satellite link measurements to generate BDS-3 precise orbits and the orbit predictions. The orbit prediction is used to form the broadcast precise orbit parameters. The 1-h orbit predictions are at the same level accuracy as estimated orbits. Then the inter-satellite link measurements are collected real-time and used to estimate precise clock offsets every minute. BDS-3 onboard clocks have perfect short term frequency stability and the error caused by the 1-min update interval is negligible.

The strategy of precise orbit determination of BDS-3 satellites by combining code and phase measurements at 7 ground stations distributed in Chinese mainland and inter-satellite link (ISL) measurements is given in Table 7 (Tang et al., 2018; Yang et al., 2019).

The MGEX CODE Product (Montenbruck et al., 2013) is chosen as reference to evaluate the orbit accuracy. Root mean square (RMS) values of orbit differences for all satellites are shown in Fig. 4, and the mean RMS is summarized in Table 8.

By processing inter-satellite link measurements, the differences of clock offsets between two satellites are obtained. By joint processing all the clock offsets differences with least squares adjustment, the redundant observations are used to remove measurement noise of the ISL facilities and thus high accuracy inter-satellite clock offsets are obtained.

Clock offset accuracy from MGEX CODE products and ISL measurements is evaluated using RMS of detrended residuals of clock offsets. The results demonstrate that ISL clock offsets generally have a smaller residual than CODE clock offsets, with average values of 0.16 ns and 0.21 ns for all satellites, respectively. Table 9 provides the detailed information on clock offsets, including differences between the two methods on March 2, 2022.

Illustrative examples of detrended clock offset variations between CODE products and ISLs are provided in Fig. 5 for satellites C26 and C28 relative to C38. C26 exhibits almost identical variations between the two approaches, while slight differences are observed for C28. The results suggest that ISL measurements only without ground stations may offer precise clock offsets with better accuracy than the CODE products which are estimated using hundreds of ground stations.

The results confirm that by combining regional code and phase measurements with inter-satellite link measurements, the orbit and clock offsets of BDS-3 satellites with centimeter accuracy can be obtained real-time, making precise positioning available to global users.

Advancement implementation

The two abovementioned technological advancements can be used to overcome the convergence problem and the regional coverage problem respectively. The key issue of the abovementioned technological advancements in BDS-3 is how to implement these in current BDS-3 frame i.e., the 3GEO + 3IGSO + 24MEO constellation. The advancements can be implemented within either the hourly-updated broadcast message of positioning, navigation, and timing service from IGSO and MEO satellites or the PPP-B2b signal from GEO satellites.

From the evaluation by Montenbruck et al., 2020, the broadcast clock parameter errors are more than 0.30 m

Table 6 Positioning performance of BDS PPP-AR/PPP-RTK service

	Positioning accuracy		Convergence time
	Horizontal	Vertical	
PPP-AR	< 10 cm (2 σ , 95%)	< 20 cm (2 σ , 95%)	10 min (2 σ , 95%)
PPP-RTK	< 10 cm (2 σ , 95%)	< 20 cm (2 σ , 95%)	< 3 min (2 σ , 95%)

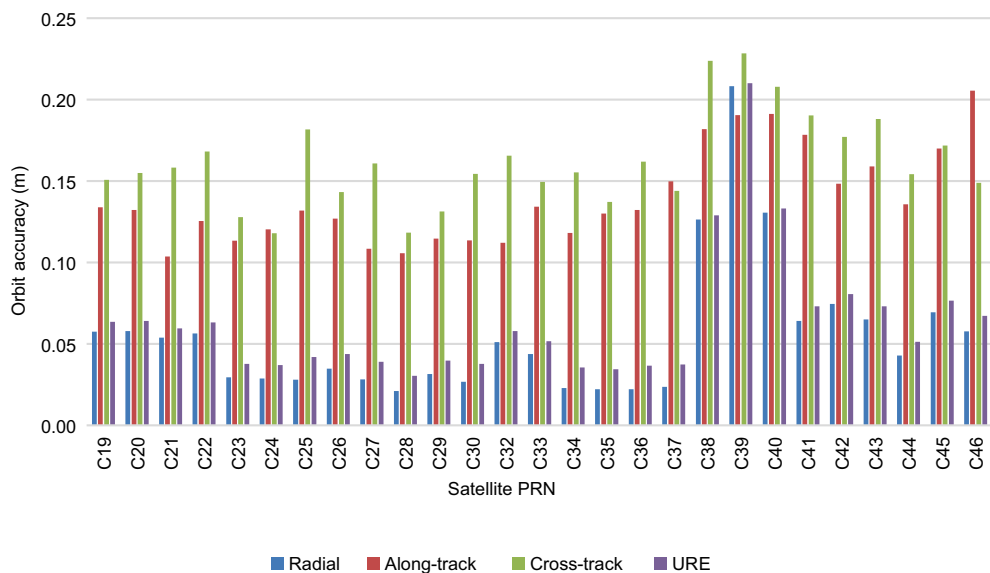


Fig. 4 BDS-3 MEO/IGSO satellite orbit accuracy compared to MGEX CODE final products. By combining the regional network and inter-satellite link measurements, the orbit URE of IGSO satellites is larger than 0.10 m and the orbit URE of MEO satellites is better than 0.05 m

Table 7 The strategy of precise orbit determination for BDS-3 satellites combining regional code and phase and ISL measurements

Observation	B1I/B3I ionosphere-free code measurements, B1I/B3I ionosphere-free phase combinations and ISL measurements
Elevation cutoff angle	7°
Data arc	72 h
Sampling rate	30 s
Weight	1: 10,000: 100 for code, phase and ISL measurements
Satellite PCO/PCV	igs14.atx
Receiver PCO	Values from manufacturers
Receiver PCV	Not applied
Station coordinates	Fixed as known values
Troposphere delay	The residual ZTD is estimated every 2 h
Solar radiation	ECOM five-parameter model
Orbit model	Two-body motion and geopotential forces from the non-spherical Earth; the attracting forces of the sun and the moon; tidal potential and solar radiation pressure

Table 8 RMS of orbit differences for different types of satellite

Satellite type	Radial (m)	Along-track (m)	Cross-track (m)	URE (m)
IGSO	0.117	0.152	0.166	0.119
MEO	0.038	0.069	0.086	0.041

and dominate the signal-in-space ranging errors of the hourly-updated broadcast message from IGSO and MEO satellites. The broadcast clock parameters are generated using Two-way Time Frequency Transfer and Inter-satellite link time comparisons (Chen et al., 2020). The broadcast clock parameter errors come from

the ranging errors of the dual-one-way time comparisons. By using the approaches in “Orbit and clock accuracy improvement with inter-satellite links” section, the orbit and clock accuracy reach centimeter level. If the approaches in “Orbit and clock accuracy improvement with inter-satellite links” section are implemented in generating the hourly-updated broadcast orbit and clock offsets, the signal-in-space errors are expected to drop from more than 0.3 m to centimeter level. Global users can get precise point positioning results with the new broadcast orbit and clock parameters. The orbit corrections of BDS-3 and GPS satellites are updated every 48 s, and the clock corrections of BDS-3 and GPS satellites are updated every 6 s through

Table 9 The STD of detrended clock offsets(ns)

PRN	CODE products	ISLs measurements	Differences between the two approaches
C39	0.17	0.15	0.11
C40	0.26	0.18	0.17
C25	0.24	0.19	0.10
C26	0.14	0.13	0.07
C27	0.23	0.15	0.14
C28	0.28	0.16	0.20
C29	0.16	0.17	0.19
C30	0.20	0.15	0.12
C19	0.14	0.11	0.13
C20	0.25	0.16	0.14
C21	0.29	0.24	0.16
C22	0.23	0.19	0.13
C23	0.18	0.14	0.14
C24	0.20	0.14	0.12
C33	0.19	0.12	0.14
C34	0.23	0.18	0.14
C35	0.15	0.13	0.12
C36	0.27	0.20	0.14
C37	0.26	0.16	0.13
C41	0.20	0.16	0.11
C42	0.21	0.17	0.12
C43	0.22	0.18	0.09
C44	0.12	0.14	0.12
C45	0.20	0.14	0.12
C46	0.29	0.19	0.14

PPP-B2b signal of GEO satellites. The clock corrections consume more than half of the message capacity of PPP-B2b. However, as evaluated by Liu et al (2022), the Allen deviations of BDS-3 and GPS satellites onboard clocks range from 3×10^{-13} to 4×10^{-13} with averaging interval of 100 s except the GPS on-board cesium clocks. Even if the clock corrections are updated every 30 s, the accuracy loss can be less than 0.005 m. Therefore, we suggest the clock corrections update interval can be expanded. The remaining message capacity is used to broadcast the uncalibrated phase delay(UPD) parameters to enable PPP-AR.

Future directions of BDS high-precision services LEO constellation enhanced precise point positioning

In recent years, a number of commercial entities, including American OneWeb, SpaceX, Boeing, Samsung of South Korea, and Aerospace Science and Technology Group of China, have publicly announced the development and deployment of their commercial low Earth orbit (LEO) constellations to enhance GNSS precise positioning performance, each consisting of hundreds to

thousands of satellites (Magan, 2015; McDowell, 2020; Meng et al., 2018). The rapid development of the LEO constellation brings innovation for precise positioning. Compared to GNSS satellites in medium Earth orbits, LEO satellites offer advantages of rapid convergence. Combining LEO navigation satellites which broadcast navigation signals to the ground receivers and GNSS satellites, is expected to speed up the convergence process of PPP and finally provide a wide-area real-time fast convergent precise positioning service. Simulation results demonstrated that LEO augmentation can significantly reduce the PPP convergence time by approximately 80% (Ge et al., 2022; Li et al., 2019). For example, by adding LEO satellites with an altitude of 1000 km, the time for a position solution to converge to 5 cm in all three directions is expected to drop within 1 min (Li et al., 2019).

The LEO satellites track BDS-3 satellites full arc and compensate for the drawback of regional network in precise orbit and clock offset determination with abundant measurements. By combining BDS-3 measurements from regional network and onboard LEO satellites, both BDS-3 and LEO orbits and clock offsets can be estimated precisely. By taking advantage of the full-arc monitoring and rapid geometry changes of LEO satellites, a global fast convergent precise positioning services will be available. Both the drawbacks of regional coverage and longer convergence can be overcome.

The large LEO constellation augmented precise point positioning service works as Fig. 6. The frame includes LEO constellation, regional monitoring network, and the ground operation center. The regional monitoring network includes the monitoring receivers tracking BDS-3 satellites and the communication facilities. The communication path of the LEO satellites and the ground operation center includes inter-satellite links, downlinks, and upload links. The LEO satellites visual in the regional network transmit the telemetry data through downlinks directly and receive orders through upload links, while the LEO satellites not visual in the regional network transmit the telemetry data and receive orders through upload links indirectly via inter-satellite link. The LEO satellites equipped with onboard BDS-3 receivers track downlink navigation signal of BDS-3 MEO satellites. The code and carrier phase measurements of LEO onboard receivers are transferred real time to the ground operation center through telemetry. The telemetry transmission paths include inter-satellite links, and downlinks. The transmission delay of the measurements onboard LEO satellites should be less than 1.0 s. The regional sparsely distributed stations track BDS-3 MEO satellites and LEO satellites and send code and carrier phase measurements to the ground operation center. The ground operation center collects the measurements from ground stations

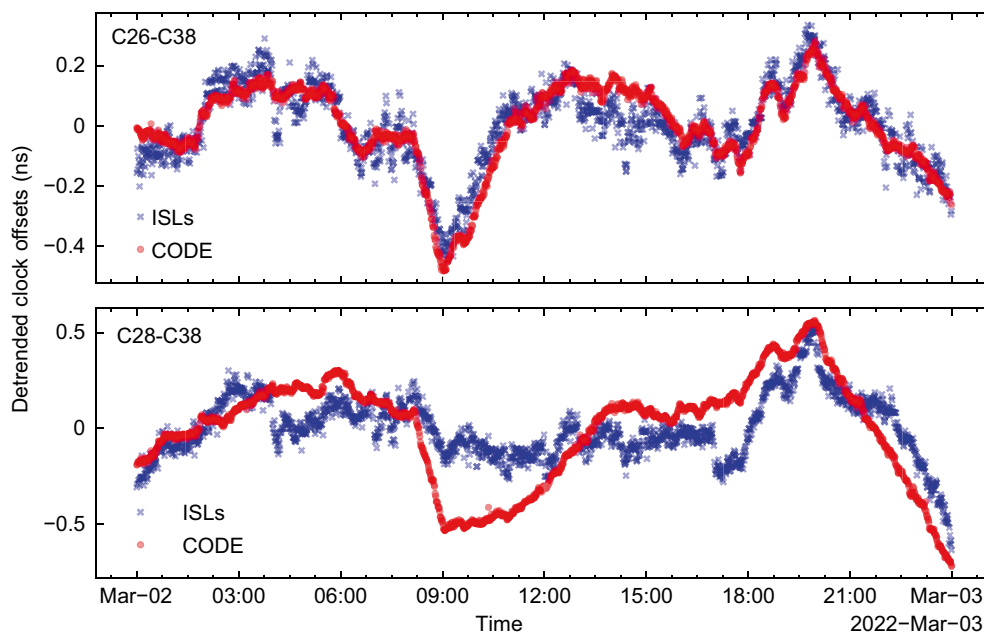


Fig. 5 Detrended clock offset variations. Illustrative examples of detrended clock offset variations between CODE products and ISLs are provided in this figure for satellites C26 and C28 relative to C38. C26 exhibits almost identical variations between the two approaches, while slight differences are observed for C28. The results suggest that ISL measurements only without ground stations may offer precise clock offsets with better accuracy than the CODE products which are estimated using hundreds of ground stations

and LEO satellites to estimate precise BDS-3 orbit and clock solutions, precise LEO orbit and clock solutions simultaneously using a real-time Kalman filter. Then the precise BDS-3 orbit and clock corrections are generated from the estimated precise BDS-3 orbit and clock solutions and uploaded to the LEO satellites. The estimated LEO orbit and clock parameters are also fitted and uploaded to the LEO satellites. The user terminals receive the broadcasted BDS-3 and LEO orbit and clock offsets and conduct the precise point positioning using BDS-3 and LEO code and phase measurements.

This section studies the feasibility of global rapidly convergent high-precision services enhanced with a LEO constellation consisting of 288 satellites. The LEO constellation consists of 288 satellites equally distributed in 24 orbit planes with an altitude of 1100 km and inclination of 55 degrees. Feasibility of precise BDS-3 and LEO satellites orbit determination is first assessed through a simulation. The simulated LEO constellation used to enhance BDS-3 orbit determination accuracy includes 18 satellites equally distributed in three orbit planes, a part of the 288 LEO constellation. By adding the 18 LEO satellites, all the BDS satellites can be observed by at least 2 receivers regardless of the ground stations or LEO satellites. Therefore, all the clock offsets can be estimated. The simulated BDS constellation includes 3 GEO satellites, 3 IGSO satellites, and 24 MEO satellites. The seven

monitoring stations are Beijing, Shanghai, Harbin, Wulumuqi, Shantou, Chengdu, and Xian. A combined orbit and clock offsets determination of BDS and LEO satellites is done with the simulated data. The estimation strategy of orbit and clock offsets is summarized in Table 10.

Accuracy of the estimated orbit and clock offsets for BDS-3 and LEO satellites are given in Fig. 7. In the figure, 01–03 in the BDS-3 constellation represent GEO satellites, 04~06 IGSO satellites, and 07~30 MEO satellites. It can be concluded that by integrating the measurements from the regional network and LEO onboard receivers, the orbit and clock offsets of IGSO satellites, MEO satellites, and LEO satellites can be estimated with centimeter accuracy. The users tracking the IGSO satellites, MEO satellites, and LEO satellites can get precise positioning results by fixing the estimated orbits and clock offsets.

The rapid convergence characterization of LEO augmented BDS precise positioning is also verified by a simulation. The simulated BDS constellation also includes 3 GEO satellites, 3 IGSO satellites, and 24 MEO satellites. The LEO constellation consists of 288 satellites equally distributed in 24 orbit planes with an altitude of 1100 km. The ground receivers simulated are located at Wulumuqi (IGS station URUM) and Beijing (IGS station BJFS). There are on average 13 LEO satellites in view. The simulated code noise is at 0.3 m level and the carrier phase noise is at 3 mm. A

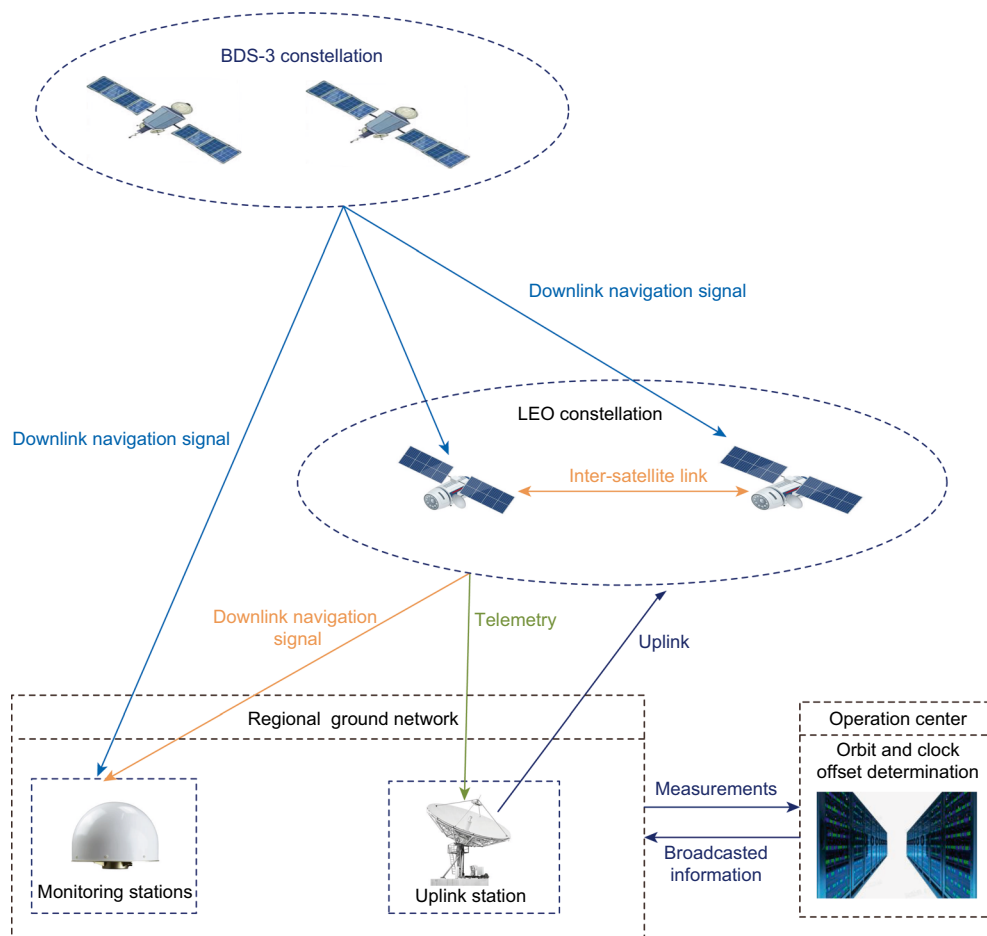


Fig. 6 Operation of large LEO constellation augmented precise point positioning service. The frame includes LEO constellation, regional monitoring network, and the ground operation center. The regional monitoring network includes the monitoring receivers tracking BDS-3 satellites and the communication facilities. The communication path of the LEO satellites and the ground operation center includes inter-satellite links, downlinks, and upload links. The LEO satellites visual in the regional network transmit the telemetry data through downlinks directly and receive orders through upload links, while the LEO satellites not visual in the regional network transmit the telemetry data and receive orders through upload links indirectly via inter-satellite link. The LEO satellites equipped with onboard BDS-3 receivers track downlink navigation signal of BDS-3 MEO satellites. The code and carrier phase measurements of LEO onboard receivers are transferred real time to the ground operation center through telemetry. The telemetry transmission paths include inter-satellite links, and downlinks. The transmission delay of the measurements onboard LEO satellites should be less than 1.0 s. The regional sparsely distributed stations track BDS-3 MEO satellites and LEO satellites and send code and carrier phase measurements to the ground operation center. The ground operation center collects the measurements from ground stations and LEO satellites to estimate precise BDS-3 orbit and clock solutions, precise LEO orbit and clock solutions simultaneously using a real-time Kalman filter. Then the precise BDS-3 orbit and clock corrections are generated from the estimated precise BDS-3 orbit and clock solutions and uploaded to the LEO satellites. The estimated LEO orbit and clock parameters are also fitted and uploaded to the LEO satellites. The user terminals receive the broadcasted BDS-3 and LEO orbit and clock offsets and conduct the precise point positioning using BDS-3 and LEO code and phase measurements

comparison of the LEO augmented BDS and BDS-only precise point positioning is given in Figs. 8 and 9. It can be concluded that with BDS-3 only it takes more than ten minutes to get positioning results better than 5 cm for station URUM and station BJFS. However, by combining BDS-3 and LEO satellites, the convergence time drops from more than ten minutes to less than 1 min for station URUM and drops from more than ten minutes to less than 1.5 min for station BJFS. By adding

LEO constellation, the convergence performance of precise positioning improves greatly.

LEO augmented PPP-RTK: ground, satellite, and LEO integrated augmentation service

Low earth orbit (LEO) satellites with an orbit altitude ranging from 300 to 1500 km have stronger signal strength and faster geometry change compared to GNSS satellites, which helps realize rapid PPP. Combining

Table 10 The strategy of combined orbit and clock offsets determination of BDS and LEO satellites

Observation	Ionosphere-free combinations from ground stations and LEO onboard receivers
Elevation cutoff angle	7°
Data arc	72 h
Sampling rate	60 s
Weight	1: 10,000 for code and phase measurements
Satellite PCO/PCV	igs14.atx
Receiver PCO	Values from manufacturers
Receiver PCV	Not applied
Station coordinates	Fixed as known values
Troposphere delay	The residual ZTD is estimated every 2 h
Clock offsets	Estimated epoch-wise
Orbit model	Two-body motion and non-spherical Earth, the attracting forces of the sun and the moon, tidal potential and solar radiation pressure for BDS-3 satellites Two-body motion and non-spherical Earth, the attracting forces of the sun and the moon, tidal potential, solar radiation pressure, atmospheric drag and empirical accelerations for LEO satellites
Estimated parameter	Initial orbit parameters, ECOM 5 solar radiation parameters for BDS-3 satellites; Initial orbit parameters, atmospheric drag factors (every 1.5 h) and empirical forces in the along-track and cross-track directions (every 3 h) for LEO satellites

PPP-RTK with low Earth orbit augmentation has the potential to achieve instant ambiguity resolution in a sparse reference station network, extend the coverage of PPP-RTK, and provide wide-area instant positioning services with an accuracy at centimeter level for massive users.

The LEO/GNSS PPP-RTK system, integrated together with LEO/GNSS PPP system, includes two components: server- and user-side. The specific processing procedure is illustrated in Fig. 10. On the server side, GNSS and LEO observations from the ground reference network are processed to generate precision satellite orbits, clocks, UPDs, and atmospheric augmentation corrections. On the client side, various users can implement GNSS/LEO convergent PPP, PPP-AR, and PPP-RTK using the various corrections provided by the server according to their own needs and scenarios.

In 2020, the BDS-3 global satellite navigation system was officially completed. Since BDS is a hybrid constellation navigation system of GEO, MEO and LEO satellites, the inclusion of low earth orbit constellation satellites is a vital step in enhancing the BDS navigation solution. To apply the LEO enhanced PPP-RTK technology to the BDS project, the receivers should be upgraded to receive the LEO satellite signals and the atmospheric corrections corresponding to the LEO satellites should also be broadcasted.

To analyze the effectiveness of PPP-RTK with LEO augmentation, the observations from the 288-LEO constellation are simulated and a regional network is employed with an average spacing of 200 km. The evaluation and

comparison of the convergence time and positioning accuracy were performed for BDS PPP (C_PPP), BDS+LEO PPP (CL_PPP), BDS+LEO PPP-AR (CL_PPP-AR), and BDS+LEO PPP-RTK (CL_PPP-RTK) solutions in this study. In PPP-RTK solution, the three reference stations are used to derive the atmospheric corrections, with accuracy better than 2 cm and 5 cm for the tropospheric and ionospheric corrections, respectively. From Fig. 11, it is revealed that the convergence time of CL_PPP-RTK was significantly shorter than C_PPP, CL_PPP, and CL_PPP-AR solutions highlighting the superior performance of CL_PPP-RTK. Moreover, the use of LEO satellites and PPP-RTK also improved the positioning accuracy by about 90% from C_PPP to CL_PPP-RTK. These results demonstrate the significant benefits of incorporating LEO satellites and PPP-RTK methods in improving the positioning accuracy and convergence time of GNSS-based applications.

In the future, with the improvement of the BDS's relevant ground facilities, research on the LEO enhanced PPP-RTK using real LEO data is expected to be carried out.

A multi-layer high-precision service frame for the future BDS

Based on the BDS high-precision services, a multi-layer high precision service frame overcoming the drawbacks of regional coverage and longer convergence time is proposed. The multi-layer high precision service frame is organized in terms of construction cost, compatibility of user terminal, operating maintenance burden, and

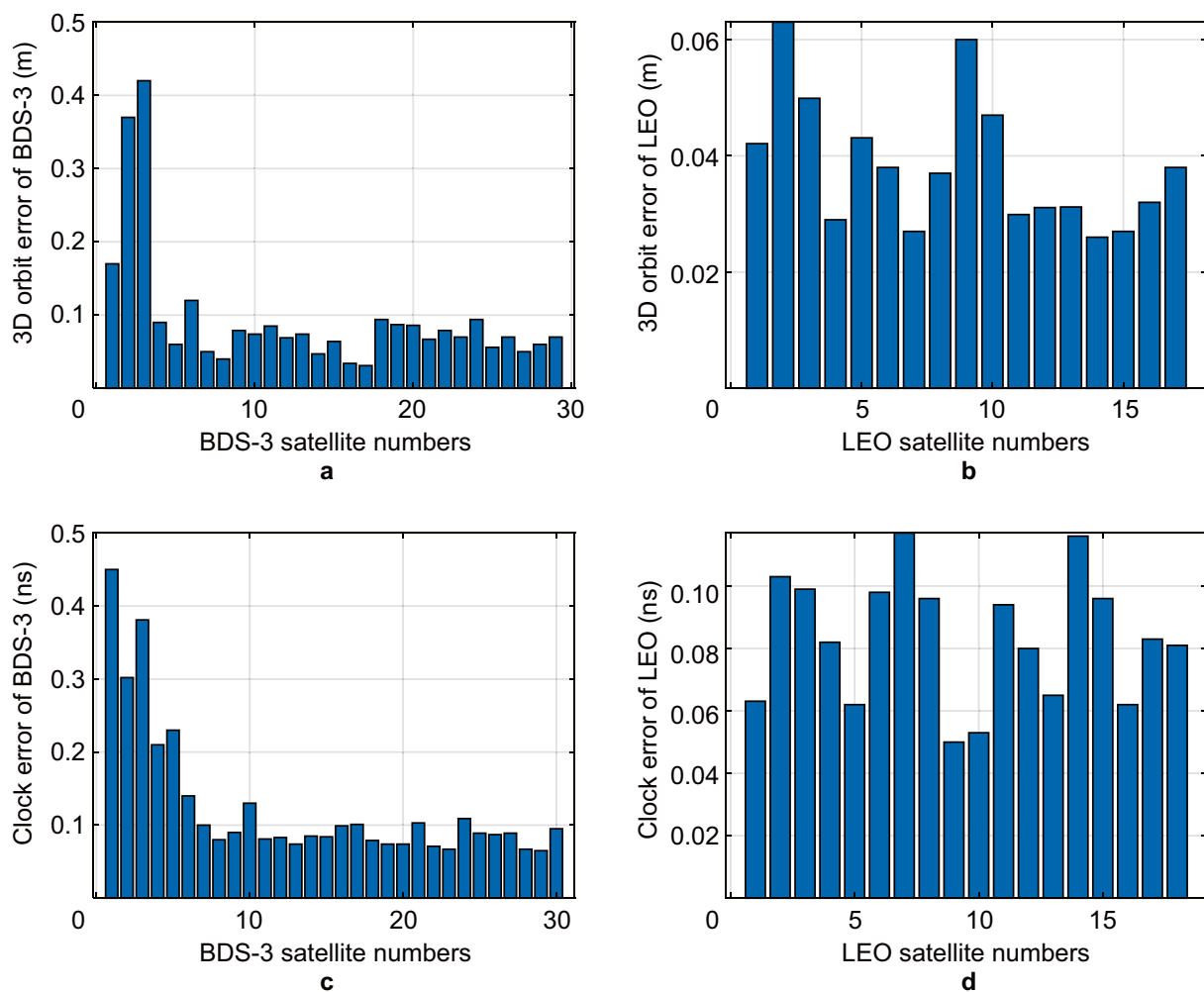


Fig. 7 BDS-3 and LEO satellites Orbit and clock offsets accuracy estimated from the combined orbit and clock offsets determination process. In the BDS-3 constellation, 01 ~ 03 represent GEO satellites, 04 ~ 06 represent IGSO satellites, 07 ~ 30 represent MEO satellites. By combined orbit and clock offsets determination of BDS and LEO satellites proposed, the LEO satellite orbits are generated with centimeter accuracy and the LEO satellite clock offsets are at 0.1 ns accuracy. The generated GEO satellite orbits are about decimeter accuracy, while the generated IGSO and MEO satellite orbits are better than 0.1 m

service level. The further BDS multi-layer High-Precision Service Frame is described in Fig. 12.

- The most fundamental layer is the global high-precision service with improved orbit and clock accuracy enhanced by inter-satellite links. Implementation of this service does not need addition or updating of any ground facilities and on-board payloads and modification of interface control document. By carrying out the approaches in “[Orbit and clock accuracy improvement with inter-satellite links](#)” section, the signal-in-space accuracy will be improved to centimeter level and high-precision positioning results will be available for global users.

This kind of service overcomes the drawback of regional coverage but still suffers from long convergence time, since it is standard PPP. Due to its low cost and compatibility with old user terminal, this kind of service is used as the most fundamental service.

- In the second layer, the regional PPP-AR and PPP-RTK are used for convergence improvement. PPP-AR can be implemented within the current GEO PPP-B2b frame by modifying the interface control document and the user terminals. However, the current BDS-3 constellation and technical status of the GEO satellites are not necessarily modified. Therefore, PPP-AR under the GEO PPP-B2b frame

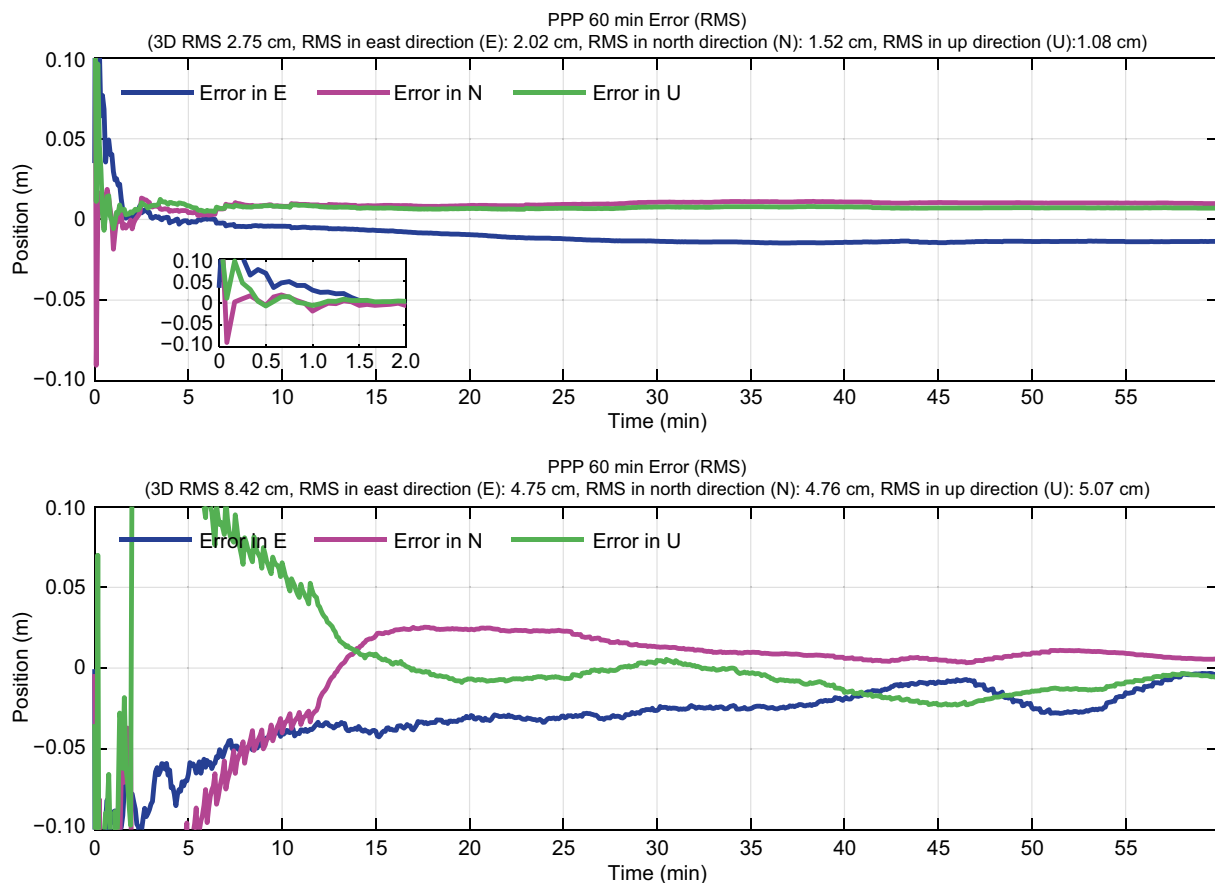


Fig. 8 Comparison of LEO augmented BDS precise point positioning errors with BDS precise point positioning errors at the first 60 min for station URUM. It can be concluded that with BDS-3 only it takes more than ten minutes to get positioning results better than 5 cm for station URUM and station BJFS. However, by combining BDS-3 and LEO satellites, the convergence time drops from more than ten minutes to less than 1 min for station URUM and drops from more than ten minutes to less than 1.5 min for station BJFS. By adding LEO constellation, the convergence performance of precise positioning improves greatly

is taken as the second layer of high-precision service. PPP-RTK is usually implemented locally due to its dependency on a tense network. The messages of PPP-RTK are usually not transmitted through satellites due to the large amount of data to be transferred.

- In the third layer, the LEO enhanced global PPP service is the one used to overcome both drawbacks of regional coverage and long convergence time. The LEO satellites play the roles in moving monitoring stations, broadcast channel, and ranging sources. The construction cost is much larger than other solutions since LEO constellation should be deployed. The operating maintenance burden is also increased since the ground operation center manages not only the MEO/IGSO/GEO satellites but also hundreds of LEO satellites. It brings new opportunities and challenges.

- LEO augmented PPP-RTK realizes instantaneous precise positioning solutions. By adding the stations of a tense network, users can realize instantaneous precise positioning solution. It is usually implemented locally.

Conclusions and discussions

BDS-3 provides PPP as a featured service, which has been made accessible to users since July 31, 2020. In this study, the accuracy of various augmentation corrections provided by PPP-B2b services and positioning performance are analyzed. The results exhibit that centimeter-level precision in both orbit and clock solutions is achievable by exploiting the PPP-B2b broadcasted information. PPP-B2b enables kinematic positioning with an accuracy at decimeter-level. However, when the performance of PPP-B2b services is compared to the world's leading augmentation services such as Galileo's HAS and QZSS's CLAS,

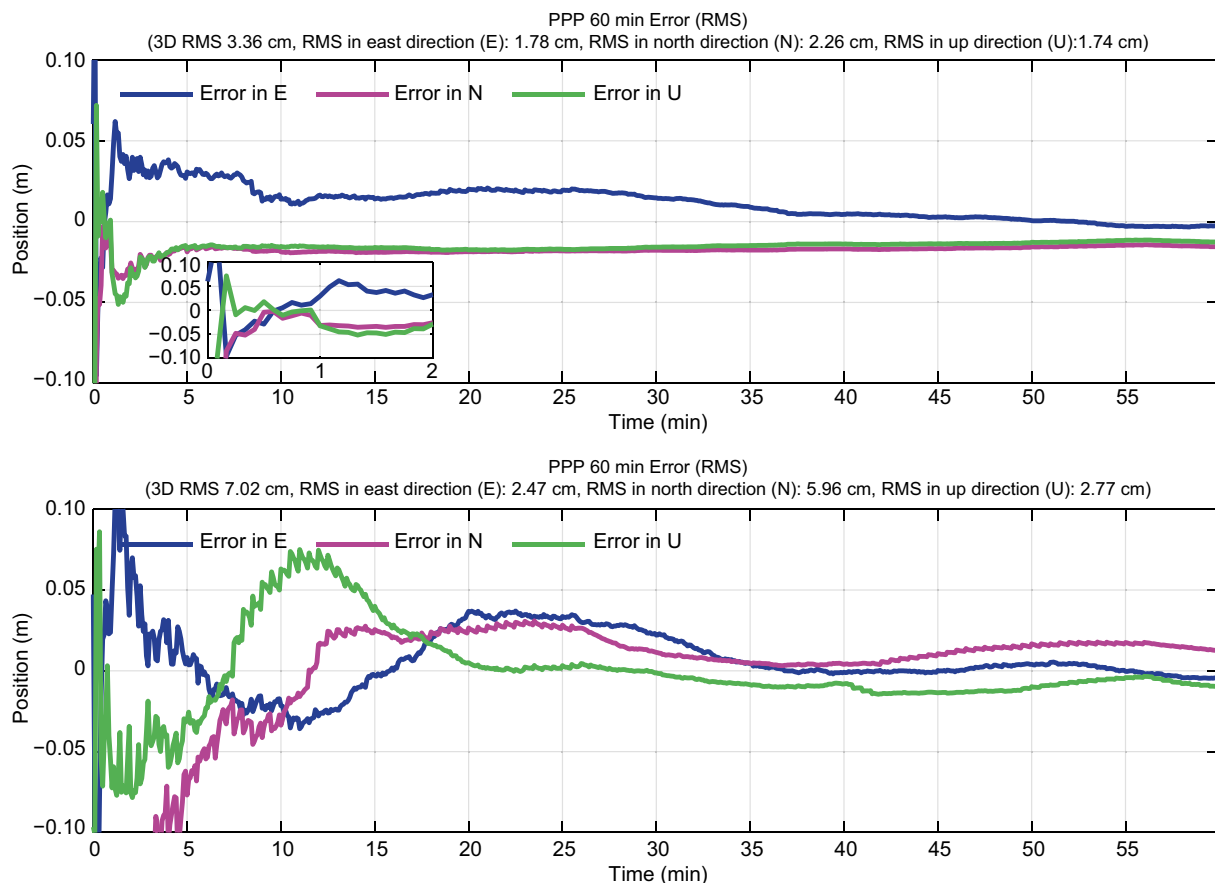


Fig. 9 Comparison of LEO augmented BDS precise point positioning errors with BDS precise point positioning errors at the first 60 min for station BJFS. It can be concluded that with BDS-3 only it takes more than ten minutes to get positioning results better than 5 cm for station URUM and station BJFS. However, by combining BDS-3 and LEO satellites, the convergence time drops from more than ten minutes to less than 1 min for station URUM and drops from more than ten minutes to less than 1.5 min for station BJFS. By adding LEO constellation, the convergence performance of precise positioning improves greatly

the BDS PPP-B2b service needs to be improved in terms of accuracy and convergence time. Furthermore, many new developments, such as autonomous driving, the Internet of Things (IoT), and smart cities, require high-precision positioning. Therefore the BeiDou high-precision navigation service should satisfy real-time, dynamic, and high-precision goals, enhancing service performance to meet the increasing expectations of users. The current PPP-B2b cannot meet all the requirements.

To further improve the performance of PPP-B2b service, this study puts forward two kinds of schemes. One can be realized in the BDS-3 frame, while the other will be realized in the future development by combining the LEO satellites.

Two approaches are proposed under BDS-3 frame. The first one is to combine PPP-RTK technology with PPP-B2b service to ensure that users can achieve PPP ambiguity resolution within minutes by adding high-precision atmospheric products to the PPP-B2b augmentation

signals. The second one is to utilize the inter-satellite links of BDS. The results demonstrate that by combining the measurements at regional monitoring stations and inter-satellite link measurements, the orbit and clock offsets of BDS-3 satellites with centimeter accuracy can be obtained throughout their full arc, making precise positioning available to global users.

In addition, the LEO constellation can also compensate for the drawback of BDS regional network. The results of the simulation experiments show that by integrating the measurements from regional networks and LEO onboard receivers, the orbit and clock offsets of IGSO, MEO, and LEO satellites can be estimated simultaneously with centimeter accuracy. Furthermore, the addition of a LEO constellation with 288 satellites significantly improves the convergence performance. Finally, the LEO augmented PPP-RTK approach is also proposed. The results show that the LEO constellation greatly improves positioning performance of the PPP-RTK solutions. In the future,

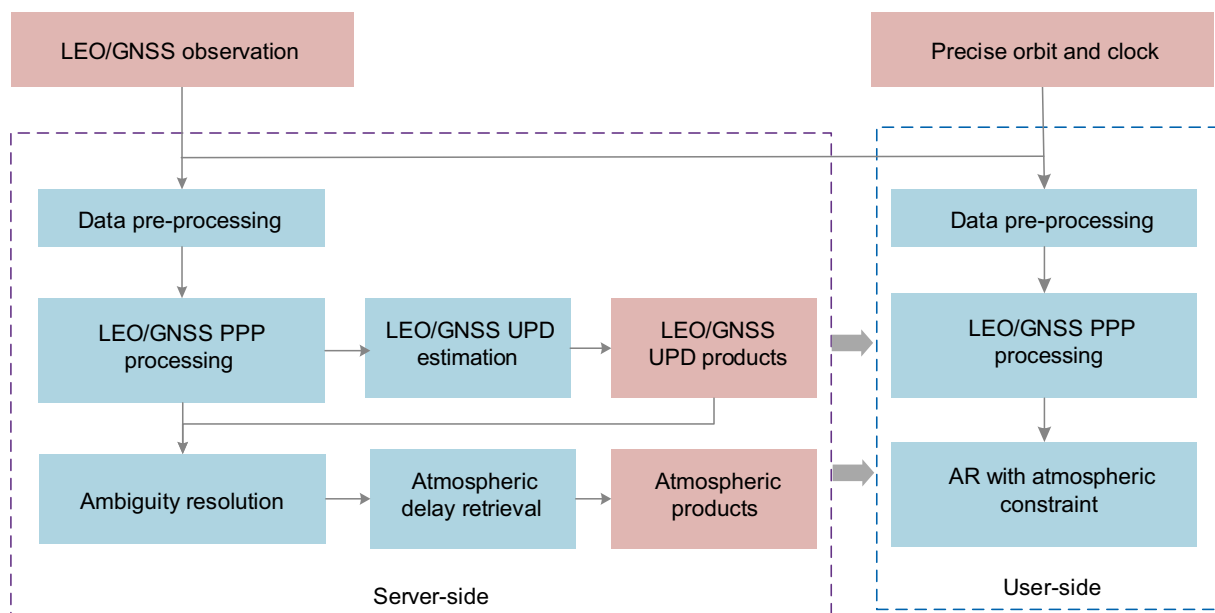


Fig. 10 System structure of LEO/GNSS integrated PPP-RTK system. On the server side, GNSS and LEO observations from the ground reference network are processed to generate precision satellite orbits, clocks, UPDs, and atmospheric augmentation corrections. On the client side, various users can implement GNSS/LEO convergent PPP, PPP-AR, and PPP-RTK using the various corrections provided by the server according to their own needs and scenarios

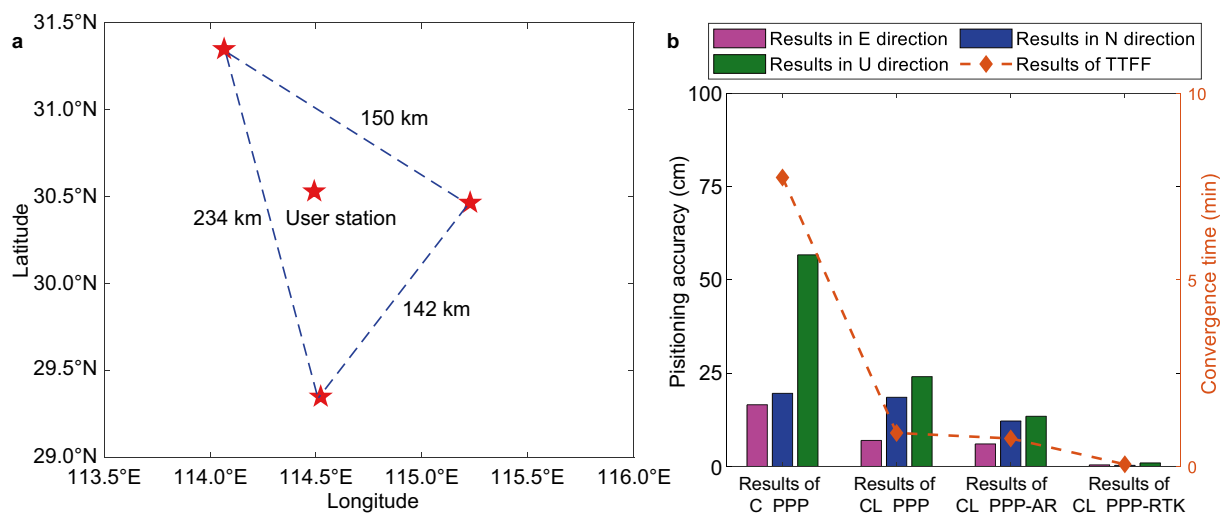


Fig. 11 The distribution of regional network (a) and positioning performance of LEO-augmented PPP-RTK (b). From this figure, it is revealed that the convergence time of CL_PPP-RTK was significantly shorter than C_PPP, CL_PPP, and CL_PPP-AR solutions highlighting the superior performance of CL_PPP-RTK. Moreover, the use of LEO satellites and PPP-RTK also improved the positioning accuracy by about 90% from C_PPP to CL_PPP-RTK. These results demonstrate the significant benefits of incorporating LEO satellites and PPP-RTK methods in improving the positioning accuracy and convergence time of GNSS-based applications

with the improvement of the BDS's relevant ground facilities, the research on LEO enhanced PPP-RTK using real LEO data should be carried out.

The above-mentioned precise positioning services are compared in Table 11. It can be concluded that precise

navigation with inter-satellite link measurements enables estimating orbit and clock offsets with centimeter accuracy and expanding the coverage area of precise point positioning service from regional to global scales. Both PPP-RTK and large LEO constellation augmented PPP

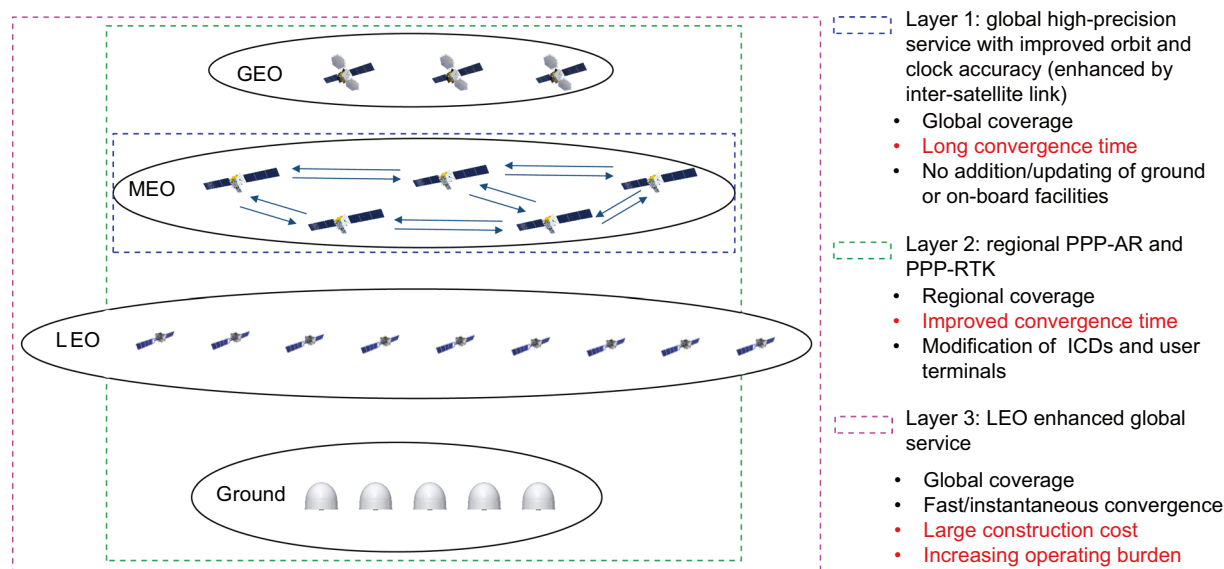


Fig. 12 The multi-layer High-Precision Service Frame for the Future BDS. The multi-layer high precision service frame is organized in terms of construction cost, compatibility of user terminal, operating maintenance burden, and service level. The most fundamental layer is the global high-precision service with improved orbit and clock accuracy enhanced by inter-satellite links. Implementation of this service does not need addition or updating of any ground facilities and on-board payloads and modification of interface control document. By carrying out the approaches in “Orbit and clock accuracy improvement with inter-satellite links” section, the signal-in-space accuracy will be improved to centimeter level and high-precision positioning results will be available for global users. This kind of service overcomes the drawback of regional coverage but still suffers from long convergence time, since it is standard PPP. Due to its low cost and compatibility with old user terminal, this kind of service is used as the most fundamental service. In the second layer, the regional PPP-AR and PPP-RTK are used for convergence improvement. PPP-AR can be implemented within the current GEO PPP-B2b frame by modifying the interface control document and the user terminals. However, the current BDS-3 constellation and technical status of the GEO satellites are not necessarily modified. Therefore, PPP-AR under the GEO PPP-B2b frame is taken as the second layer of high-precision service. PPP-RTK is usually implemented locally due to its dependency on a tense network. The messages of PPP-RTK are usually not transmitted through satellites due to the large amount of data to be transferred. In the third layer, the LEO enhanced global PPP service is the one used to overcome both drawbacks of regional coverage and long convergence time. The LEO satellites play the roles in moving monitoring stations, broadcast channel, and ranging sources. The construction cost is much larger than other solutions since LEO constellation should be deployed. The operating maintenance burden is also increased since the ground operation center manages not only the MEO/IGSO/GEO satellites but also hundreds of LEO satellites. It brings new opportunities and challenges. LEO augmented PPP-RTK realizes instantaneous precise positioning solutions. By adding the stations of a tense network, users can realize instantaneous precise positioning solution. It is usually implemented locally

Table 11 Comparison of different strategies

Service	PPP-AR	PPP-RTK	Precise navigation with BDS inter-satellite links	Navigation augmentation with 288 LEO Satellites constellation	
				large LEO constellation Augmented PPP	large LEO constellation Augmented PPP-RTK
Convergence Time	About ten minutes	about 1 min	About ten minutes	About 1 min	Instantaneous
Positioning Accuracy	Better than 10 cm	Better than 10 cm	Better than 10 cm	Better than 10 cm	
Stations Required	Globally sparsely distributed stations. More than 30 stations	Stations with range of tens of kilometers	Regional sparsely-distributed stations	Regional sparsely-distributed stations	With an average space of 200 km

reduce the convergence time greatly from tens of minutes to about 1 min. Densely distributed stations are usually required in PPP-RTK. Therefore, PPP-RTK are usually provided regionally rather than globally. It is possible to provide globally rapid convergent PPP services with regional sparsely distributed stations through large LEO constellation Augmented PPP. Also, by combining PPP-RTK and LEO constellations, users can obtain instantaneous convergent positioning result through large LEO constellation Augmented PPP-RTK.

Considering the ground station distribution requirements of the above solutions, a multi-layered high-precision navigation services architecture is proposed. Firstly, taking inter-satellite link enhanced BDS-3 precise navigation as the fundamental solution for global high-precision services. Then taking densely distributed stations augmented PPP-RTK as regional augmentations. Finally, taking LEO constellation enhanced PPP services as the increment for the future development. The multi-layered high-precision navigation services architecture is used to aid the decision making on high-precision service development.

However, the implementation may also encounter several challenges. Firstly, the widespread applications of BeiDou high-precision navigation services could potentially strain the system, necessitating effective management and optimization of service capabilities to ensure stable operations. Secondly, with a more complex augmented service framework, establishing a comprehensive integrity monitoring system becomes imperative to safeguard the reliability of the services. Overall, the prospects for BeiDou high-precision services are promising, but continual adaptation and resolution of new challenges are essential to maintain its leading position in the field of precision navigation.

Abbreviations

BDS	BeiDou navigation satellite system
GEO	Geostationary earth orbit
IGSO	Inclined geostationary earth orbit
MEO	Medium earth orbit
LEO	Low earth orbit
SISRE	Signal-in-space ranging error
GBAS	Ground based augmentation system
SBAS	Satellite based augmentation system
PPP	Precise point positioning
PPP-AR	Precise point positioning with ambiguity resolution
PPP-RTK	Precise point positioning real-time kinematic

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Not applicable.

Author contributions

Weiguang Gao and Wei Zhou proposed the idea, evaluated service performance of PPP-B2b and wrote the paper. Xingxing Li and Yongqiang Yuan conducted LEO augmented positioning and PPP-RTK experiments. Chengpan Tang and Xiaogong Hu conducted ISL enhanced orbit and clock estimation experiments. Weiguang Gao and Wei Zhou conducted the LEO augmented BDS orbit determination experiments.

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Availability of data and materials

The derived result data is available under request.

Declarations

Competing interests

Xingxing Li is an editorial board member for *Satellite Navigation* and was not involved in the editorial review or decision to publish this article. All authors declare that they have no competing interests.

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